Department of Electrical & Electronic Engineering
Brac University
Semester: Spring 25

EEE101L

ELECTRICAL CIRCUITS I LABORATORY

Section:05

Software Experiment 04

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Group Number: 05
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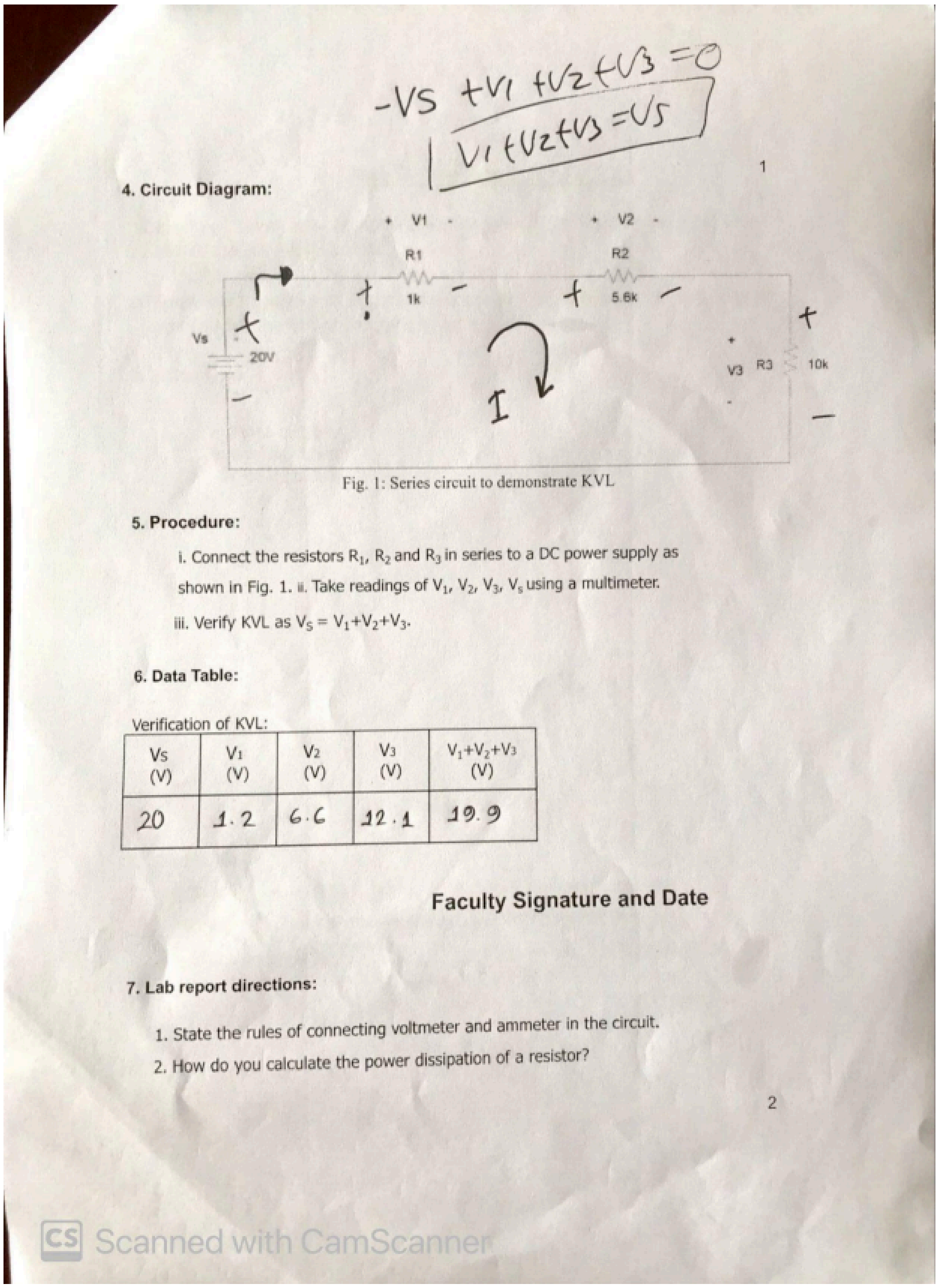
Objective:

Verification of KVL and KCL, Introduction to Series-Parallel Circuits, Verification of Superposition Principle & Thevenin's Theorem in PSpice.

Verification of KVL and KCL

Part-A: Verification of KVL



$-V_S + V_1 + V_2 + V_3 = 0$

$\boxed{V_1 + V_2 + V_3 = V_S}$

1

4. Circuit Diagram:

Fig. 1: Series circuit to demonstrate KVL

5. Procedure:

i. Connect the resistors R_1, R_2 and R_3 in series to a DC power supply as shown in Fig. 1. ii. Take readings of V_1, V_2, V_3, V_s using a multimeter.

iii. Verify KVL as $V_S = V_1+V_2+V_3$.

6. Data Table:

Verification of KVL:

Vs (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1+V_2+V_3$ (V)
20	1.2	6.6	12.1	19.9

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7. Lab report directions:

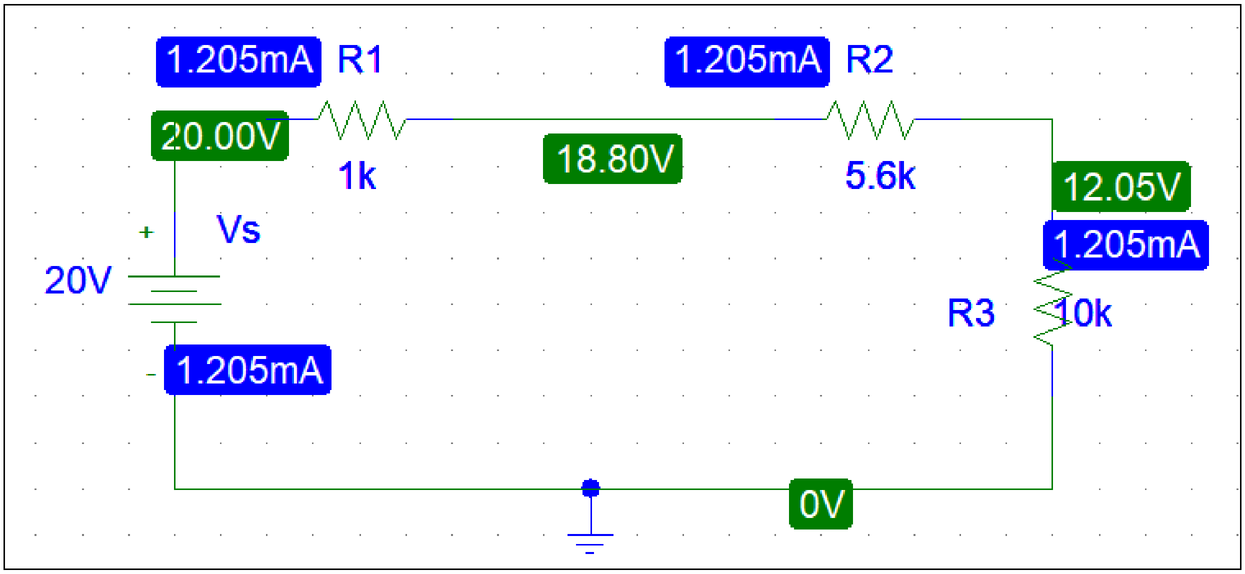
1. State the rules of connecting voltmeter and ammeter in the circuit.
2. How do you calculate the power dissipation of a resistor?

2

Lab Report Part -A

Here,

$\therefore$ Error = {(20 - 19.90) / 20 } × 100 % = 0.5 %



Here, Vs= 20V

V1=(20-18.80)=1.2V

V2=(18.80-12.05)=6.75V

V3=(12.05-0)=12.05V

Vs	V1	V2	V3	V1+V2+V3
20	1.2	6.75	12.05	20

Here,

Error = 0 %

Part-B: Verification of KCL

Part-B: Verification of KCL

1. Objective: This experiment is intended to verify Kirchhoff's current law (KCL) with the help of a series parallel circuit.

2. Theoretical Background: KCL states that the algebraic sum of the currents entering any node equals the sum of the currents leaving the node.

3. Equipment:

- Three resistors
- One multimeter
- One DC supply

4. Circuit Diagram:

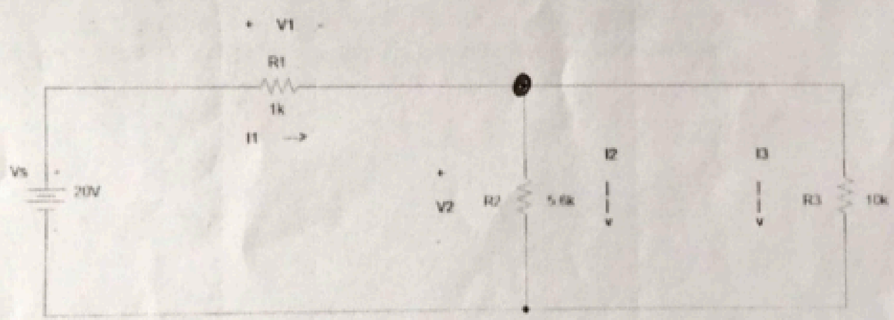
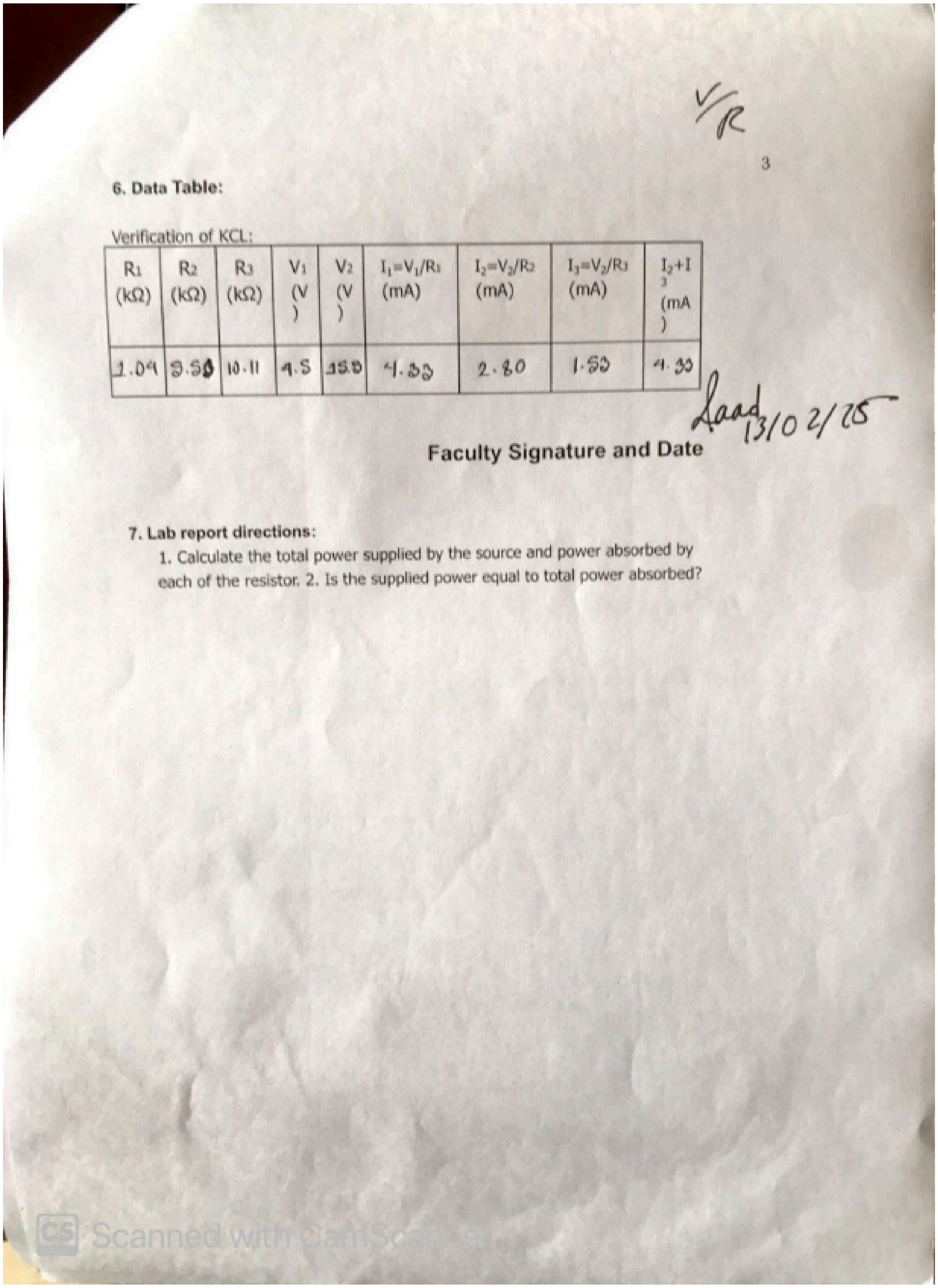


Fig. 2: Verification of KCL

5. Procedure:

i. Connect the resistors in series-parallel across the power supply as shown in Fig. 2. ii. Measure the voltage drop across the resistors.

iii. Calculate the currents I_1, I_2 and I_3 by using $I_1 = V_1/R_1$, $I_2 = V_2/R_2$ and $I_3 = V_2/R_3$ as given above and prove that $I_1 = I_2+I_3$.



V/R

3

6. Data Table:

Verification of KCL:

R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	V_1 (V)	V_2 (V)	$I_1=V_1/R_1$ (mA)	$I_2=V_2/R_2$ (mA)	$I_3=V_2/R_3$ (mA)	I_2+I_3 (mA)
1.04	3.50	10.11	4.5	15.3	4.33	2.80	1.53	4.33

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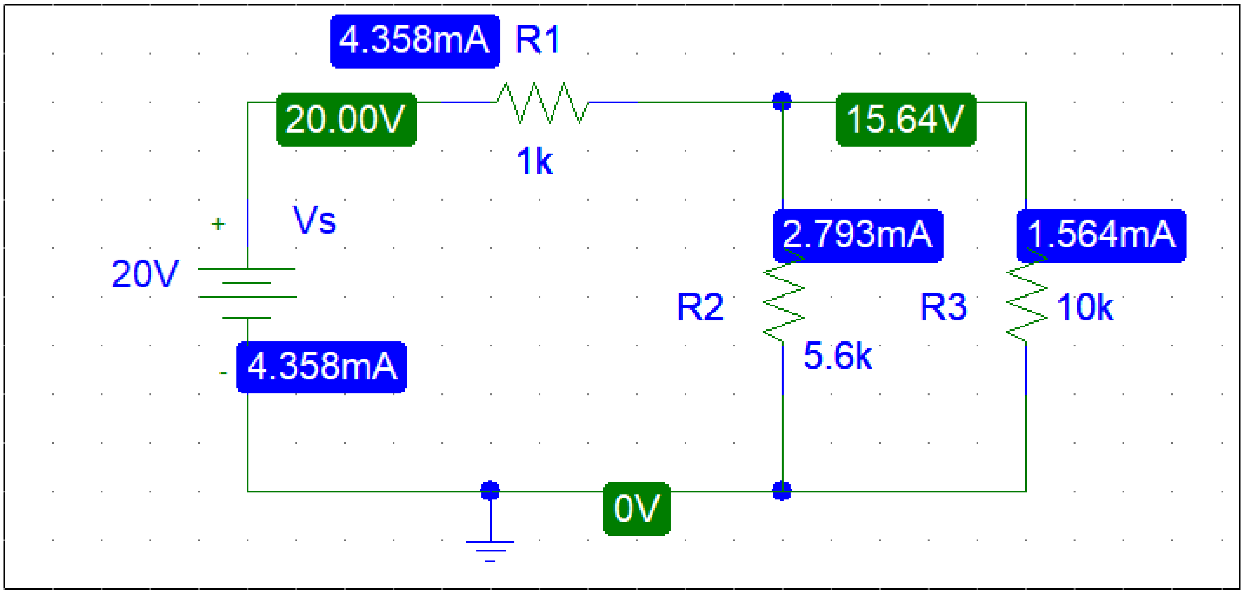
7. Lab report directions:

1. Calculate the total power supplied by the source and power absorbed by each of the resistor. 2. Is the supplied power equal to total power absorbed?

Lab Report Part -B

Here,

∴ Error = { I1 - (I2 + I3) / I1 } × 100 % = { 4.33 - (4.33) / 4.33 } × 100 % = 0%



R1	R2	R3	V1	V2	I1(mA)	I2(mA)	I3(mA)	I2+I3 (mA)
1k	5.6k	10k	4.36v	15.64	4.358mA	2.793mA	1.564mA	4.357mA

Here,

ERROR= { I1 - (I2 + I3) / I1 } × 100 %

= {4.358-(2.793+1.564)/4.358}× 100 %

= 0.02294%

ERROR is changed ..

Introduction to Series-Parallel Circuits

4. Circuit Diagram:

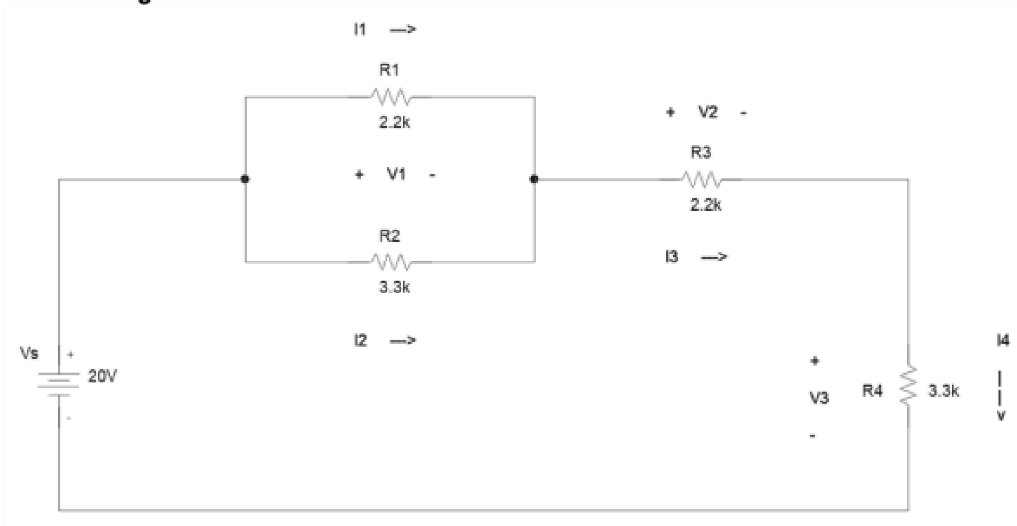


Fig. 1

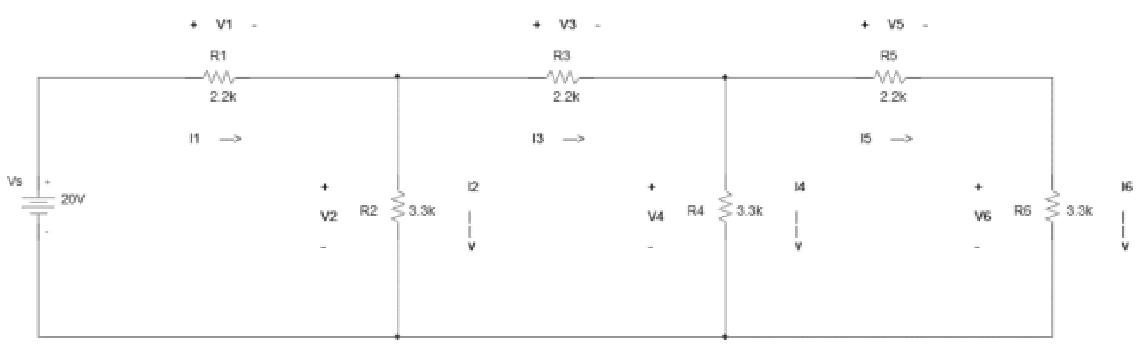


Fig. 2

5. Procedure:

i. Set up the circuit as in the above figures.

ii. Measure the voltage across the resistors and calculate the currents.

Lab Report

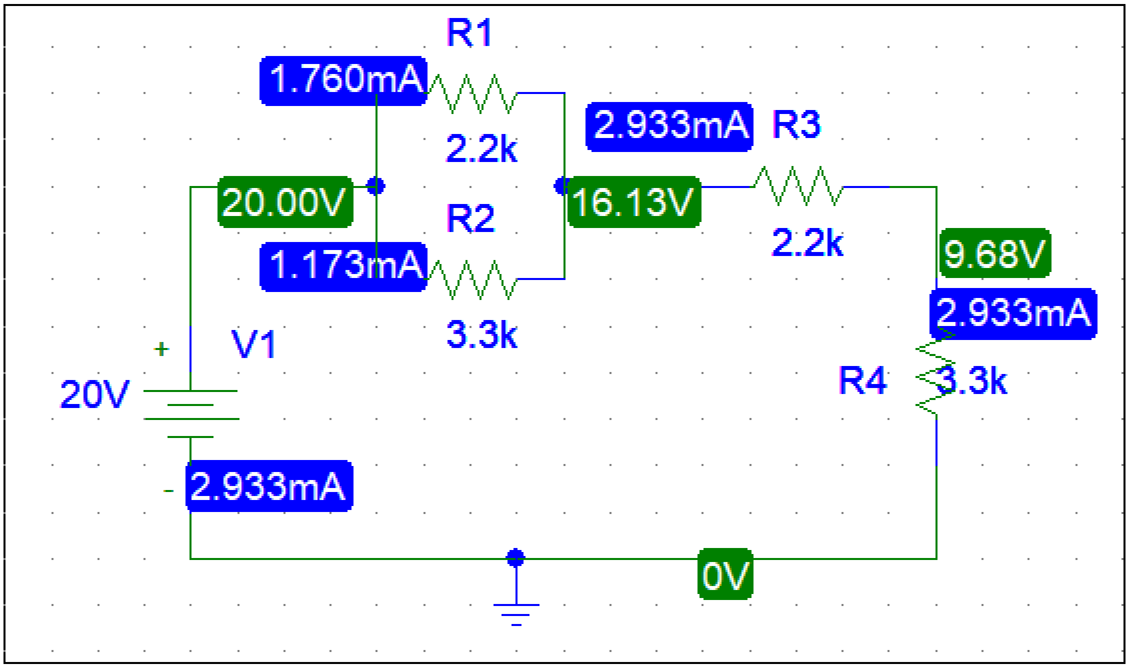


Fig-1

R1	R2	R3	R4
2.2k	3.3k	2.2k	3.3k

V1	V2	V3
3.87V	6.45V	9.68V

I1	I2	I3	I4
1.760mA	1.173mA	2.933mA	2.933mA

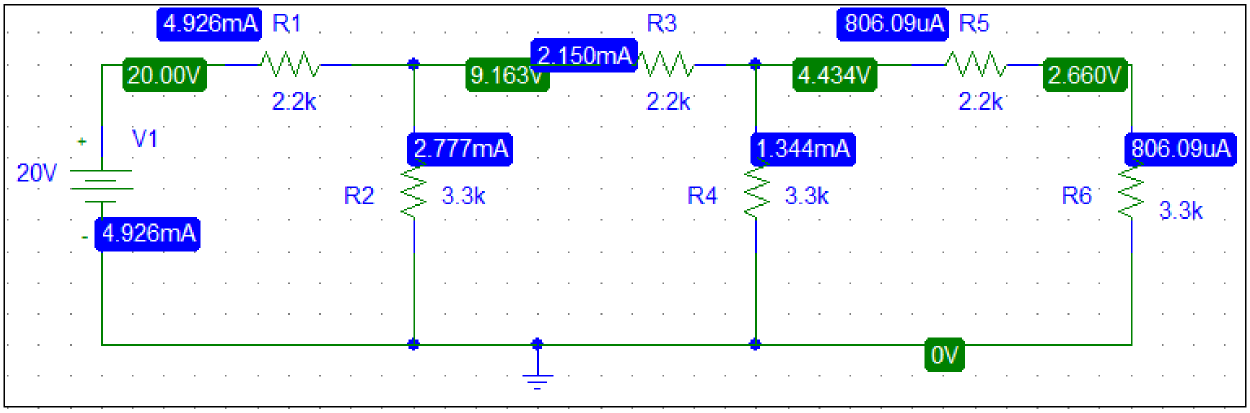


Fig-2

R1	R2	R3	R4	R5	R6
2.2k	3.3k	2.2k	3.3k	2.2k	3.3k

V1	V2	V3	V4	V5	V6
10.837V	9.163V	4.729V	4.434V	1.774V	2.660V

I1	I2	I3	I4	I5	I6
4.926A	2.777A	2.150A	1.344A	0.806A	0.806A

So, In figure 1,2 simulation Report we can see that current and voltage are little bit changed from the experimental report.

Verification of Superposition Principle

4. Circuit Diagram:

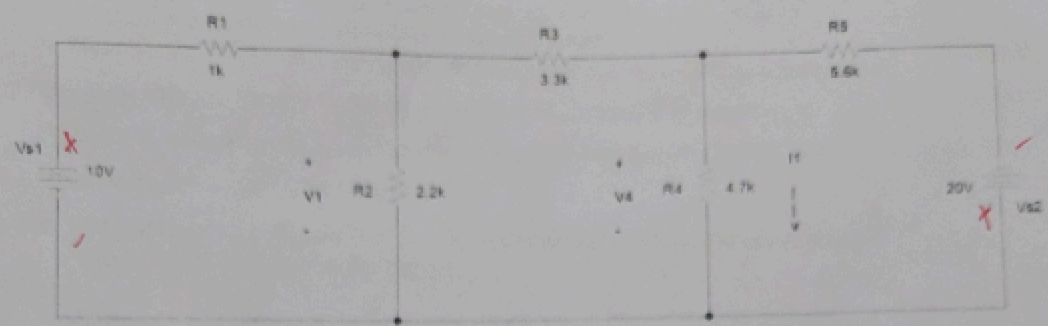


Fig. 1

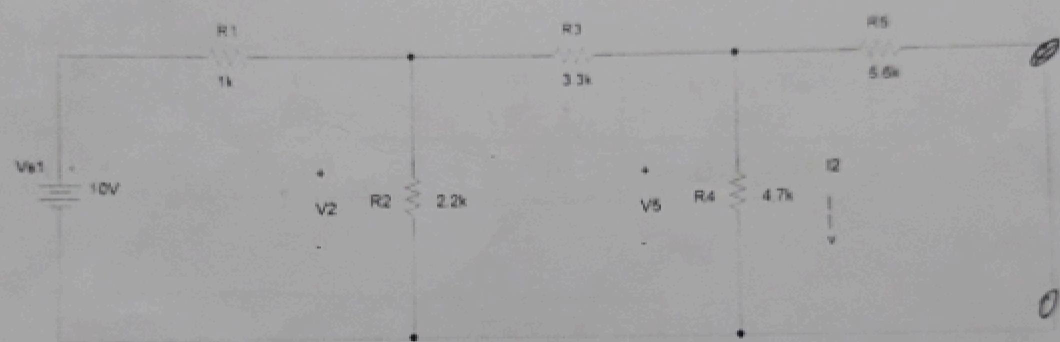


Fig. 2

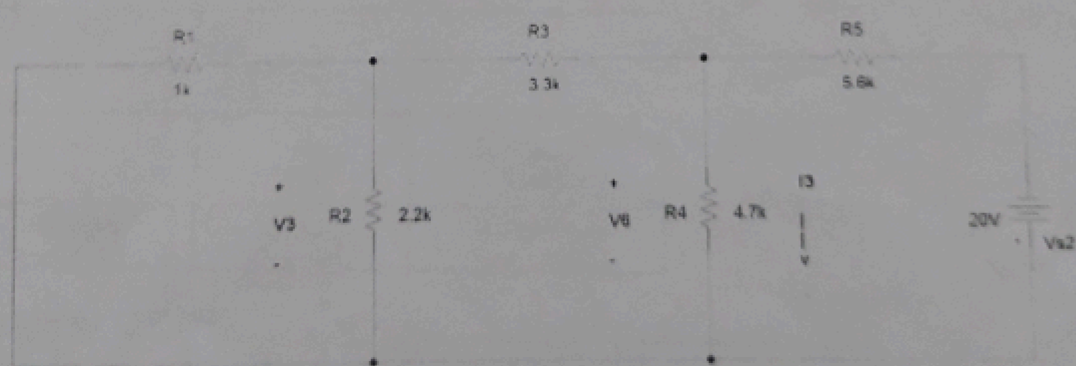


Fig. 3

5. Procedure:

i. Setup the circuit as in Fig. 1.
ii. Measure resistor values accurately with DMM.

iii. Apply V_{s1} = 10 volts from one source and V_{s2} = 20 volts from another source. Measure voltages accurately with DMM.

iv. Measure V_1 with DMM. Measure V_4 and calculate the current I_1 and record it in the given table. v. Render V_{s2} inactive (keeping V_{s1} active) and measure V_2. Measure V_5 and calculate I_2. vi. Render V_{s1} inactive (keeping V_{s2} active) and measure V_3. Measure V_6 and calculate I_3.

vii. Verify if $V_1 = V_2 + V_3$ and $I_1 = I_2 + I_3$ which would validate the superposition theorem for this particular circuit.

6. Data Table:

Fig. 1

R4 (kΩ)	V1 (V)	V4 (V)	$I_1 = V_4 / R_4$ (mA)
4.7	5.15	-2.95	-0.6276

Fig. 2

V2 (V)	V5 (V)	$I_2 = V_5 / R_4$ (mA)
6.05	2.62	0.5574

Fig. 3

V3 (V)	V6 (V)	$I_3 = V_6 / R_4$ (mA)
~~2.51~~ -0.96	~~6.30~~ -5.63	~~1.3404~~ -1.1936

Verification

V1 (V)	$V_2 + V_3$ (V)	I1 (mA)	$I_2 + I_3$ (mA)
5.15	~~8.55~~ 5.07	-0.6276	-0.6362

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Lab Report

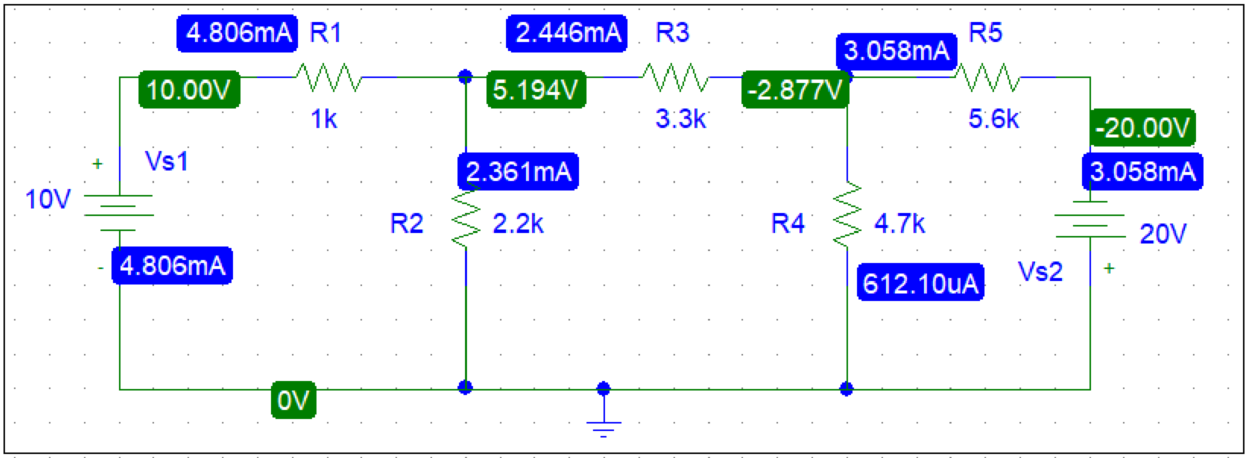

Fig-1

Here,V1 is R2’s Voltage and V4 R4’s Voltage.

R4	V1	V4	I1=V4/R4
4.7k	5.194V	-2.877V	-0.612A

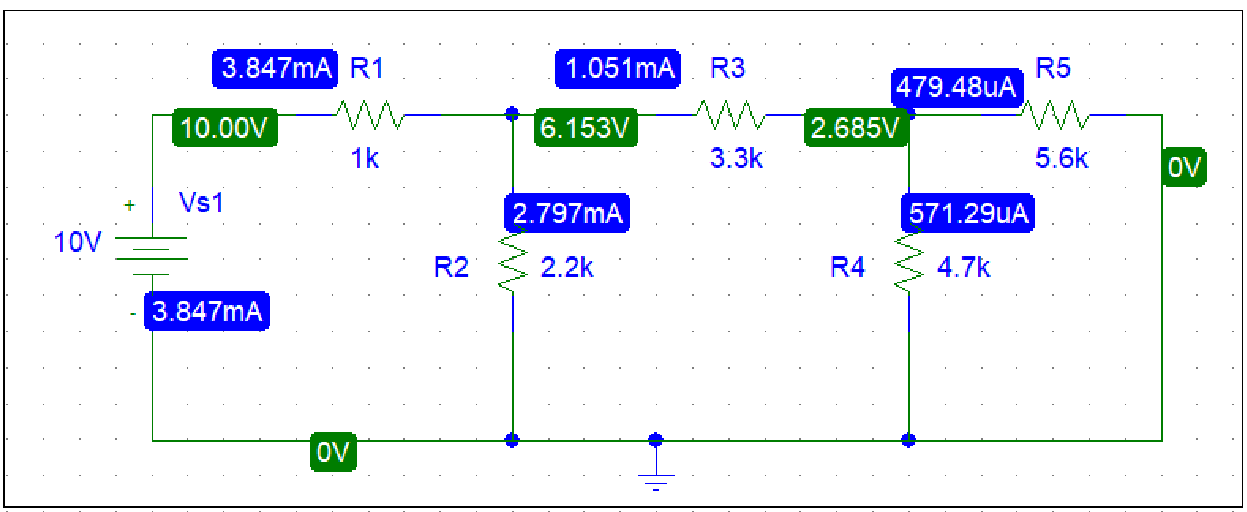

Fig-2

Here,V2 is R2’s Voltage and V5 R4’s Voltage.

R4	V2	V5	I2=V5/R4
4.7k	6.153V	2.685V	0.571A

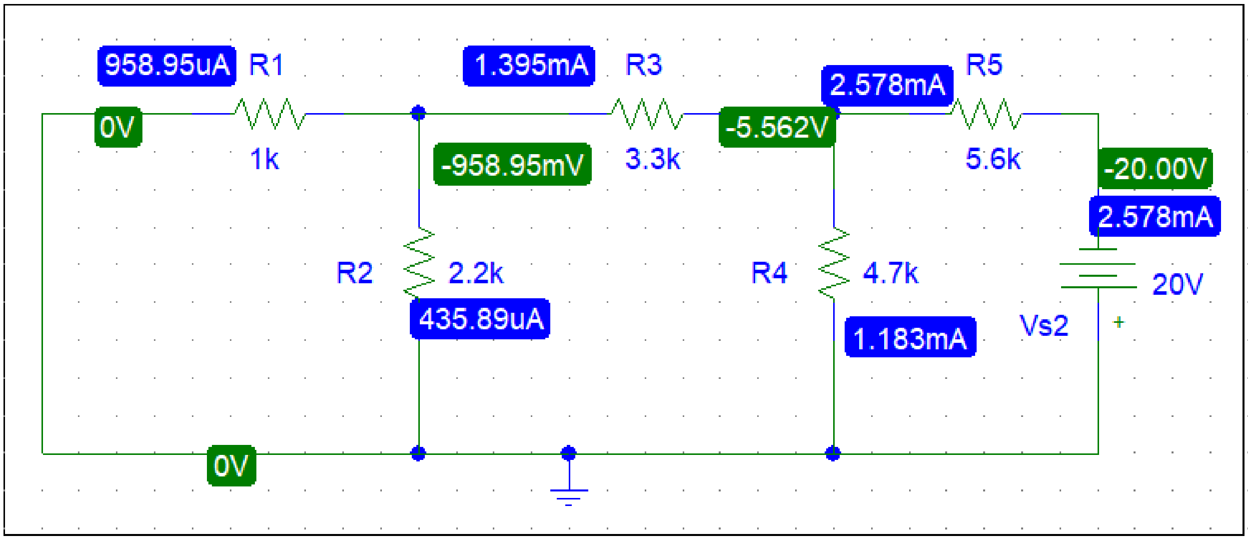

Fig-3

Here,V3 is R2's Voltage and V6 R4's Voltage.

R4	V3	V6	I3=V6/R4
4.7k	-0.958V	-5.562V	-1.183A

Verification

V1	V2+V3	I1	I2+I3
5.194V	5.195V	-0.612A	-0.612

Here is a comparison between experimental value and simulation value.

Conclusion:

In this experiment, we successfully verified fundamental circuit theorems and concepts using PSpice simulations. We confirmed Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) with minimal error, demonstrating accurate voltage and current relationships in closed loops and at circuit nodes.We also explored series-parallel circuits, where the simulation results showed small deviations in voltage and current values, likely due to simulation rounding or modeling assumptions. The observed data aligned closely with theoretical predictions, confirming proper understanding of complex resistive networks. Furthermore, the Superposition Principle and Thevenin's Theorem were validated by comparing partial and combined source effects. The current and voltage calculations across resistors confirmed that the individual source effects add linearly, reinforcing our grasp of linear circuit behavior. Overall, this experiment enhanced our practical understanding of foundational circuit laws and analysis techniques, and it demonstrated the effectiveness of simulation tools like PSpice in visualizing and validating theoretical concepts.